MV-GenomSeq → MII KDS mapping: issues to address

9 October 2026

Reference: Thomas Debertshäuser's annotated mapping table `DRAFT-Mapping-Table-genomDE-DK-2026-10-06-TD-final.xlsx`. Row references are Excel row numbers in his tabs: ***O*** *= KDK oncology,* ***R*** *= KDK rare diseases,* ***G*** *= GRZ,* ***OI*** *= Open Issues. Every item below was checked against the real MII packages (2026 pins and the 2027.0.0 ballot) and cross-reviewed; details in `docs/REVIEW-TD-2026-10-06.md`.*

1. Summary

	Items	Meaning
Fix in our table now	22	wrong against 2026 **and** 2027; apply against the pinned 2026 packages
Ballot comments (≈ 27 Oct 2026)	17	no standardised MII representation in 2027
Already filed by Thomas (HDB-975/976)	4	support and align
MV-GenomSeq specification	12	ConceptMaps, local code systems, assembly rules
2027-only changes to prepare	8	mapper changes at the package switch
Disagreements with the notes	8	to discuss
Rows without a note, still open	8	

2. Fix in our mapping table now

1. **MolGen canonical URLs.** The real URLs are `…/modul-molgen/StructureDefinition/variante` and `…/diagnostische-implikation`; `mii-pr-molgen-kopienzahlvariante` and `-strukturvariante` do not exist (MolGen has one generic variant profile; MTB has CNV and DNA/RNA-fusion profiles, no generic SV profile). 57 rows carry invalid canonicals. *Rows:* R 18–19, 61–115; OI 64–75. *Also:* priorRds[].diagnosticDate/.diagnosticResult belong to `…/molekulargenetischer-befundbericht` — R 18–19.
2. **Stale Genomics-Reporting slice names** (`dna-change`, `amino-acid-change`, `genomic-dna-change`, `genomic-ref/alt-allele`, `variant-type`, `transcript-ref-seq`) → `representative-coding-hgvs` 48004-6, `representative-protein-hgvs` 48005-3, `genomic-hgvs` 81290-9, `ref-allele` 69547-8, `alt-allele` 69551-0, `coding-change-type` 48019-4, `representative-transcript-ref-seq` 51958-7. 28 rows. *Rows:* O 134, 154, 156, 164, 166–167, 169–176, 190; OI 34 | R 64, 69, 74, 78, 83–84, 89, 92, 94, 96, 103, 107, 112; OI 69.
3. **RD CNV/SV rows → generic `variante`** with components `chromosome-identifier`, `exact-start-end`, `allelic-state`, `gene-studied`, `cytogenomic-nomenclature`; CNV `type` gain/loss → `coding-change-type` SO:0001742 / SO:0001743. *Rows:* R 61–79, 100–115; OI 64–67, 72–75 (notes R 62–65, 67–71, 74, 77–79, 81–82, 101–104, 106–109, 112, 115).
4. **ECOG: `unknown` → obds code `U`, never `dataAbsentReason`** (value[x] and `coding:obds` are 1..1). `5` and `notApplicable` stay unrepresentable (ballot, §3). Reverses our earlier NUM-OMICS resolution. *Rows:* O 10, 76.
5. **RD ACMG class → `diagnostische-implikation.component:clinical-significance`** (53037-8; binding is *example*, LL4034-6 recommended), one implication per variant via `derivedFrom:variant`. *Rows:* R 61, 80, 100.
6. **RD mode of inheritance → `diagnostische-implikation.component:mode-of-inheritance`**, HPO (HP:0000006/7, HP:0001417, HP:0001427; *preferred* binding, so our LOINC codes validate but HPO is what MII uses). `unclear`: omit or keep a text value. *Rows:* R 73, 93, 111.
7. **RD segregation → `variante.component:variant-inheritance`** 94186-4 (LL5489-1, extensible): deNovo LA26807-0, fromFather LA26321-2, fromMother LA26320-4; `fromMotherAndFather` needs an own `biparental` code (HDB-976 c); `notPerformed` has no faithful code (ballot). *Rows:* R 76, 97, 114; OI 67, 71, 75.

8. **RD publications** → `diagnostische-implikation.extension:workflow-relatedArtifact` (RelatedArtifact, PMID as url/citation). *Rows:* R 75, 95, 113; OI 66, 70, 74.
9. **RD ACMG criteria: no target in MolGen 2026/2027** → DRAFT, ballot (§3). *Rows:* R 62–63, 81–82, 101–102.
10. **Localization (oncology and RD)** → `mii-pr-molgen-molekulare-konsequenz.component:feature-consequence` (SO; all codes verified inside the value set): RD codingRegion SO:0001580, splicingRegion SO:0001630, regulatoryRegion SO:0001566, intronic SO:0001627, intergenic SO:0001628; oncology coding SO:0001580, inRegulatoryElements SO:0001566, notInCoding… lossy. Interim until HDB-976. *Rows:* O 142, 164, 188; R 72, 92, 110; OI 29, 34, 44, 65, 69, 73.
11. **Oncology prior diagnostics → MTB container** `mii-pr-mtb-ngs-bericht` linked from `behandlungsepisode.supportingInfo:Vorbefund`; variants as `result:EinfacheVariante`; `date` → `effectiveDateTime`; `type` → `genomic-study-analysis.extension:method-type` (karyotyping/array/other exist via the GR code system; `single`, `none` do not); complexVariants[] is free text → `DiagnosticReport.conclusion`. *Rows:* O 41–57; OI 5–21.
12. **MTB plan.** Follow-up therapy `reference` → `Procedure.extension:causedBy` (exists since 2026.0.1); `priority` → `extension:Prioritaet.valuePositiveInt` (not `MedicationRequest.priority`); therapy variants[] → `reasonReference:StuetzendeMolekulareAlteration` plus mandatory `reasonReference:Primaertumor`; study variants[] → `ServiceRequest.supportingInfo` (interim); therapy `evidenceLevel(+Details)` → `extension:Evidenzgraduierung` (`Z(FDA)` has no code); recommendedStudies rows split ServiceRequest vs ResearchStudy. *Rows:* O 90, 206–207, 211, 218–220, 230; OI 25, 49–54, 57; sibling row O 222.
13. `therapyResponseSource` **is a data source**, not the RECIST/RANO method → `response-befund.note/derivedFrom`; `method` 1..1 has no genomDE source (ballot). *Rows:* O 100; OI 26.
14. `additionalClassification` → `mii-pr-onko-weitere-klassifikationen` (exists since 2026.0.3): `system` → `code.text`, `key` → `value.text` (+ preferred codings) → MAPPED. *Rows:* O 2–3; OI 2–3.
15. `therapeuticStrategy` **interim** → `therapieempfehlung-kombination.RequestGroup.code` (preferred binding, `mii-cs-onko-therapie-typ`) or example-bound `MedicationRequest.category`; standard binding via ballot. *Rows:* O 228; OI 55.
16. **Fusions:** `geneA`/`geneB` → `five-prime-gene`/`three-prime-gene`, profile by `sequenceType` (dna-/rna-fusion); mandatory breakpoint components emitted with `dataAbsentReason` (their values are 0..1); `structureType` → `coding-change-type` (SO) competes with MII's own MolGen example (gene_fusion in the same 0..1 slot) → question 7. *Rows:* O 178–185, 189–190; OI 36–43, 45.
17. **MTB molecular.** Ploidy → MAPPED (NCIt C18303, `valueQuantity`; no `interpretation` against 2026 — its value set is missing in GR 3.0.0); CNV type → `component:cnv-type` (bwhc code system, unbound) plus `coding-change-type`; expression → `valueCodeableConcept` present/absent + `interpretation` H/L/N (interim); SBS signatures → COSMIC URI `https://cancer.sanger.ac.uk/cosmic` with normalised code `SBS6`, catalogue version in `coding.version`; LOH → open component on `einfache-variante` (interim). *Rows:* O 130, 134, 144, 151–153, 165; OI 28, 30, 32–33, 35.
18. **Seltene strategy:** code system is 1:1 to the enum (`genetic` = gene-therapy), but split by value set: medication strategies → `MedicationRequest.category:MVGenomSeqTherapieStrategie`, non-medication → `ServiceRequest` (nicht-medikamentös). Our `Therapieempfehlung.Type` target was wrong. *Rows:* R 129.
19. **Consent domains → DRAFT.** OIDs .2/.6/.8 exist but mean IDAT/MDAT purposes; `provision.provision.code` is required-bound, so mvSequencing, reIdentification, caseIdentification have no conformant form. Our mapper emits the literal enum strings — a hard conformance failure. *Rows:* O 112; R 46; G 49; OI 27, 63, 82.
20. **GRZ devices:** `Procedure.device` does not exist in R4 → `Procedure.extension:device` (GR `genomic-study-analysis-device`: `extension:device` → Device, `extension:function`). *Rows:* G 5–6, 8–9, 11–12, 16–19, 38–39.
21. **GRZ QC:** `sequencingLayout` → qc `sequencing-read-type`; `minCoverage` is **not** `sequencing-coverage` (GR: % of regions covered) → `metrics-description` interim; same for fragmentation, non-coding flag, bases-above-quality. The qc extension's context is declared as Extension, not Procedure (ballot). Rows G 29, 35 already use the extension — align. *Rows:* G 7, 30–33, 40; OI 76–81.
22. **Metachrone diagnoses (boolean):** no element; derive from follow-up Conditions with `occurredFollowing` → first diagnosis (2027 only, see §6), keep the flag when no diagnosis accompanies it. *Rows:* O 79; OI 22.

3. Ballot comments (deadline about 27 October 2026)

#	Module	Issue	Rows	Ask	Status
B1	Onkologie	ECOG `5` / `notApplicable`	O 10, 76	add `5` + NA to the ECOG code system (and ConceptMap), or relax value[x]/`coding:obds` to 0..1	to file
B2	Onkologie	TNM `coding:uicc` 1..1 (new in 2027)	O 32–36	allow `coding:uicc` 0..1 when `coding:snomed-ct` is present; UICC↔SNOMED maps incomplete (38 clinical / 39 pathological of 131 unmatched)	to file
B3	Onko / MTB	therapy type on recommendations	O 228; OI 55	`category` slice on `therapieempfehlung(-medikation)` bound to `mii-vs-onko-systemische-therapie-art`	to file
B4	Onkologie	MV-GenomSeq artefact defects	—	`mii-cm-onkologie-to-mvgenomseq`: TNM rows → `tnmClassifications`, `TNMSchluessel` LOINC/SNOMED mix-up, empty Condition map; LM blocks unmapped	to file
B5	Onkologie	hereditary cases	O 9, 87–88, 197–198; OI 4, 23–24, 47–48	`diagnosticAssessment` (5-state, no element), preventive-measure result/reference, therapeutic/risk-reducing flags (extend `mii-ex-onko-operation-intention`); purpose of the unused 2027 predisposition value sets	to file
B6	MTB	transcript binding	O 49–52, 169–172; OI 13–16	representative-transcript-ref-seq.value[x] required to `http://ensembl.org` (not a ValueSet; blocks RefSeq) → example/extensible like MolGen	to file
B7	MTB	`coding-change-type` 0..1	O 53–56, 173–176; OI 17–20	0..* or document multiple codings per CodeableConcept	to file
B8	MTB	prior findings	O 41–42, 57; OI 5–6, 21	profiled free-text carrier; effective[x] MS on `ngs-bericht`; profile the `ngs-bericht` → `genomic-study` link; add single-gene and `none` method codes	to file
B9	MTB	response assessment	O 100; OI 26	`response-befund.method` 1..1 (RECIST/RANO) → 0..1 or `other`/`unknown`; place for the data source	to file
B10	MTB	study recommendations	O 90, 206–207, 218–220; OI 25, 49–50, 52–54	add ServiceRequest to the Evidenzgraduierung context **and** slice it on `studieneinschluss-anfrage`; `Z(FDA)` code; `reasonReference` → Observation; `systemische-therapie.basedOn` → `studieneinschluss-anfrage`	to file
B11	MTB	recommendation type	O 229; OI 56	inLabel / offLabel / compassionateUse / secPreventive / unknown: no element on Therapieempfehlung	to file
B12	MTB	CNV / expression / LOH / SBS	O 134, 144, 151–153, 165; OI 30, 32–33, 35	define CNV-type code system (+ genomDE 6 levels); coded expression level; `loh` concept + component; SBS-signature biomarker concept	to file

#	Module	Issue	Rows	Ask	Status
B13	MTB	fusions	O 178–185, 189; OI 36–43, 45	relax 1..1 breakpoint components; HGNC binding on partner-gene values; profile for intragenic SVs	to file
B14	MTB	QC extension / defects	O 130; G 7, 30–33, 40; OI 28, 76–81	fix qc `context`; clarify sub-extension semantics (mean vs minimum coverage); add min-coverage, bases-above-Q, fragmentation; allele-A/B displays swapped; NCIt system URI in ploidie	to file
B15	MolGen	HDB-976 (localization profile + VS, Befundkategorie CS, `biparental`)	O 142, 164, 188; R 66, 72, 76, 86, 92, 97, 105, 110, 114; OI 29, 34, 44, 64–65, 67–69, 71–73, 75	**support**; add: ACMG criteria component; variant-inheritance.value[x] 1..1 blocks not-performed; study design single/duo/trio (R 20; OI 59)	filed by TD + addenda
B16	Seltene	HDB-975 (GMFCS, hospitalisation bands)	R 6, 22–23, 29; OI 58, 60–62	**support**; add `other` to the medication-strategy value set (R 129)	filed by TD + addendum
B17	Consent	MV-GenomSeq domains	O 112; R 46; G 49; OI 27, 63, 82	policy codes for mvSequencing, reIdentification, caseIdentification under OID …`24.5.3`	to file

4. MV-GenomSeq specification items

#	Item	Rows
S1	ConceptMap localization → SO (oncology 3 values, RD 5)	O 142, 164, 188; R 72, 92, 110; OI 29, 34, 44, 65, 69, 73
S2	ConceptMap TNM SNOMED → UICC (inverse of MII maps, coverage audited)	O 32–36
S3	ConceptMap SV `structureType` → SO	O 190
S4	ConceptMap ECOG → LOINC (2027 required binding)	O 10, 76
S5	Local code system CNV type (6 levels)	O 134
S6	Local code system Befundkategorie (interim for HDB-976)	R 66, 86, 105; OI 64, 68, 72
S7	Local code system recommendation use-type	O 229; OI 56
S8	Local code systems hospitalisation bands (two)	R 22–23; OI 60–61
S9	COSMIC NamingSystem for SBS signatures	O 151–153; OI 32–33
S10	`biparental` code, study-design code system	R 76, 97, 114; OI 67, 71, 75; R 20; OI 59
S11	Derived rules: metachrone flag; `sequenceType` → fusion profile; one implication per variant	O 79, 189; R 61, 80, 100; OI 22, 45
S12	Assembly rules: Patient per donor, Specimen → donor, report → result, identifier systems, Encounter vs EpisodeOfCare; feed `mii-lm-mvgenomseq-onkologie` / `mii-cm-onkologie-to-mvgenomseq` from our table	—

5. Where we disagree with the notes

Rows	Note says	Package says
O 112; R 46; G 49; OI 27, 63, 82	OIDs .8/.2/.6 exist in no version	they exist in every version (IDAT erheben, MDAT erheben, MDAT wissenschaftlich nutzen) and MII's own examples use them for Broad Consent; the ballot conclusion still holds because they mean the wrong thing
O 90; OI 25	link only via `CarePlan.activity`	`Procedure.extension:causedBy` → `mii-pr-mtb-therapieempfehlung` exists since 2026.0.1
O 100; OI 26	`Observation.method` RECIST/RANO	field is "Source of therapy response data"; `method` 1..1 has no genomDE source
O 57; OI 21	no concept for karyotyping/array/other	the value set includes the whole GR code system; only single-gene and `none` are missing
O 79; OI 22	further Condition	it is a boolean flag; the 2027 `occurredFollowing` target is the real news
G 7, 30–33, 40; OI 76–81	`minCoverage` → `sequencing-coverage`	GR semantics: % regions covered; `read-depth` = mean depth; no slot for a minimum
R 73, 93, 111	LOINC codes apply only to the MTB counterpart	MTB uses the same preferred HPO binding; LOINC is not invalid in either
O 2–3, 41–57, 130, 134, 144, 151–153, 165, 178–185, 189–190; OI 2–3, 5–21, 28, 30, 32–33, 35–43, 45	"2027"	identical in the 2026 packages except the interpretation value set and closed value[x] slicing; nothing to wait for

6. 2027-only changes to prepare for the package switch

#	Change	Rows
C1	TNM: `coding:uicc` 1..1 and `coding:snomed-ct` 0..1 slices; value[x].coding 1..*; y/r/a `modifierExtensions` (c/p `cpPraefix` already in 2026)	O 32–36
C2	ECOG: `coding:loinc` gets a required binding (LA9622-7…LA9627-6); `focus` → primary tumour	O 10, 76
C3	Diagnosis: `category:onkologie` 55342001 1..1; `bodySite.coding:icd-o-3` required-bound (licensed terminology, not on tx.fhir.org); morphology extension 0..*; `fruehere-tumorerkrankung` needs two categories	O 17–20, 27–31, 37–40
C4	`occurredFollowing` may target `diagnose-primaertumor` (basis of the metachrone rule); `transformation-von` extension added	O 79; OI 22
C5	MTB interpretation value set `mii-vs-mtb-interpretation-high-low` (H/L) on ploidie, hrd-score, mutationslast, brcaness	O 130; OI 28
C6	value[x] slicing closed on `rna-seq` and `einfache-variante`	O 144; OI 30; variant rows above
C7	QC sub-extension values typed (Quantity / string / CodeableConcept)	G 7, 30–33, 40; OI 76–81
C8	Panel DeviceDefinition `mii-pr-mtb-panel-devicedefinition` for kit names (via `Device.definition`); Onko variant profile switches HGNC to `http://www.genenames.org/geneId` (MTB/MolGen keep `http://www.genenames.org`)	G 5–6, 8–9, 11–12, 16–19, 38–39

7. Rows without a note that are still open

Rows	Field	Status
O 9; OI 4	`diagnosticAssessment`	no element in Onko/MTB (not the LM's Keimbahndiagnose) → ballot B5

Rows	Field	Status
O 87–88; OI 23–24	preventive-measure result / reference	no element → ballot B5; interim ServiceRequest + Procedure.basedOn
O 150; OI 31	expression reference cohort	→ `rna-seq.note` (spec)
O 194; OI 46	other recommendations	mappable now: `CarePlan.activity:obds.detail.code` WW/AS/WS/OP/ST/KW/SO
O 197–198; OI 47–48	intervention flags	no element → ballot B5
R 20; OI 59	study design single/duo/trio	no concept → MolGen ballot (B15)

8. Questions for Thomas

1. Who files the ballot comments in §3 — we or you? Can we see the HDB-975/976 texts?
2. Consent: who registers MV-GenomSeq policy codes under OID …`24.5.3` (AG Consent/TMF), and when?
3. ECOG `5`/`notApplicable`: extend the code system or relax value[x]? Follow-up ECOG: only `effectiveDateTime` discriminates?
4. TNM: inverse ConceptMap acceptable for the mandatory UICC coding; who audits SNOMED → UICC coverage; can `coding:uicc` become 0..1?
5. Transcript binding `http://ensembl.org`: intentional? Which system URIs for Ensembl / RefSeq?
6. HGVS strings: KDK gives bare `c./g./p.V600E`; prefix and normalise? (GR `hgvs-vs` has no filter.)
7. `coding-change-type` 0..1: several SO codings in one CodeableConcept? On fusions: gene_fusion vs structural mechanism?
8. Mandatory value[x] components: omit for `unclear`/`notPerformed` (lossy), own codes, or relax to 0..1?
9. Localization: MTB `result` slice for `molekulare-konsequenz`, or reference the MolGen profile? Fallback for `notInCodingAndNotInRegulatoryElements`?
10. `therapeuticStrategy`: wrap every recommendation in `therapieempfehlung-kombination`, or ballot a `category` slice?
11. SBS signatures: COSMIC NamingSystem URI and owner (MII vs MV-GenomSeq IG)?
12. GRZ QC: one disease-agnostic model (MolGen GR metrics) or per module (MTB qc)? Is the 2027 panel DeviceDefinition catalogue the home for GRZ kit names?
13. Use our table to fill `mii-lm-mvgenomseq-onkologie` / `mii-cm-onkologie-to-mvgenomseq`?
14. Identity and linkage rules for MV-GenomSeq bundles; which HGNC system URI (`http://www.genenames.org` vs …`/geneId`)?
15. ICD-O-3 topography becomes a required binding in 2027: which terminology server should implementers validate against?